

rect() — four numbers, two jobs

```
rect(x, y, width, height);
```

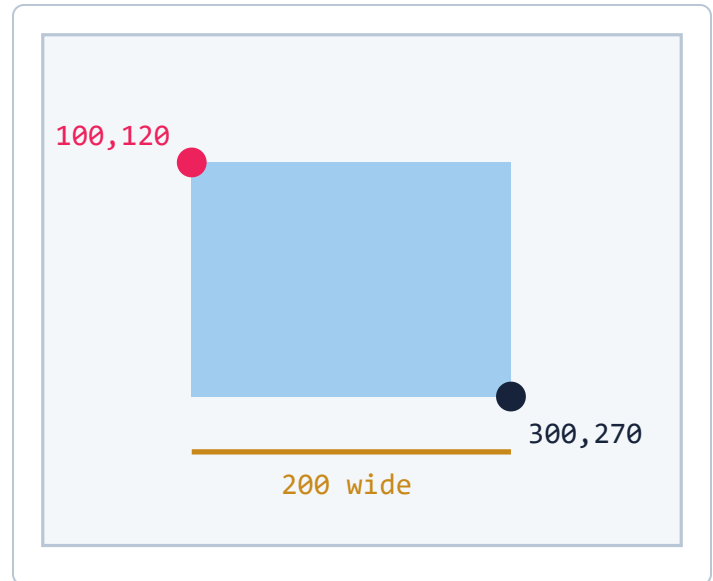
The first two say **where**. The last two say **how big**. They are not another corner.

To find the edges, add:

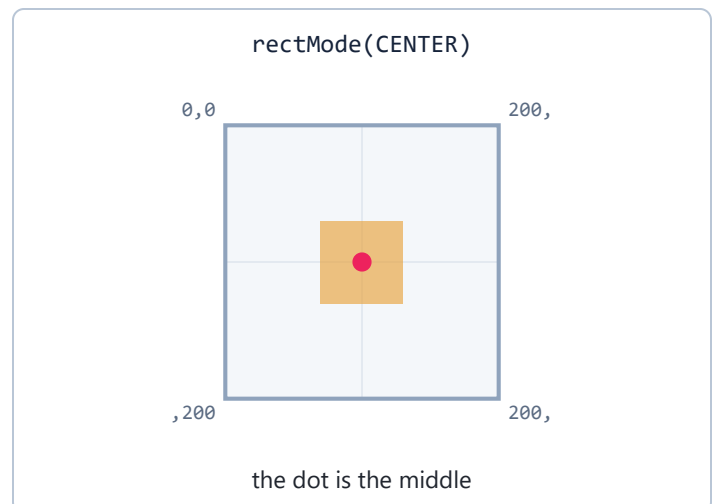
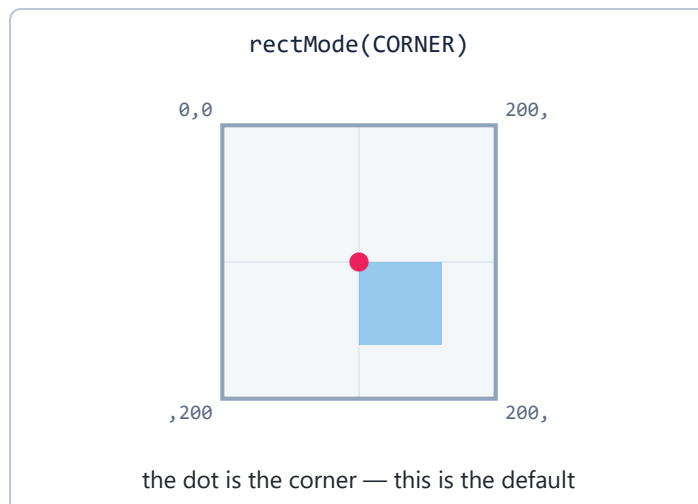
```
rect(100, 120, 200, 150)
```

right edge = $100 + 200 = 300$

bottom edge = $120 + 150 = 270$



rectMode() — what the first two numbers mean



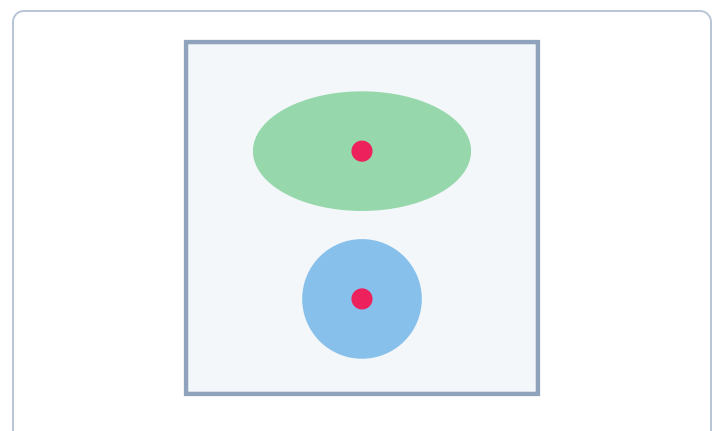
Both pictures are `rect(100, 100, 60, 60)`. Only the mode changed. Put `rectMode(CENTER);` in `setup()` once — every `rect()` after it obeys.

Round shapes

```
ellipse(x, y, width, height);
circle(x, y, diameter);
```

Same four numbers as `rect`, but **x, y is the center**. `circle` is the shortcut for when width and height are equal.

Ellipses have a mode too, and it is already `CENTER` — which is why they have always behaved the way you expected.



Other things you already know

CALL	WHAT IT DOES
<code>triangle(x1,y1, x2,y2, x3,y3)</code>	three corners, closes itself
<code>line(x1,y1, x2,y2)</code>	two points, start and end
<code>fill(gray)</code>	one number, 0–255
<code>fill(r, g, b)</code>	three numbers, each 0–255
<code>noStroke()</code>	no outline on shapes
<code>background(r, g, b)</code>	repaints the whole canvas

`fill()` and `rectMode()` are **settings**. They apply to everything drawn after them, until you change them again. One `fill()` and four shapes gives you four shapes the same color.

Order matters. Later shapes cover earlier ones. Read your `draw()` top to bottom — that is the order things get painted, which is the same as distance from the viewer.